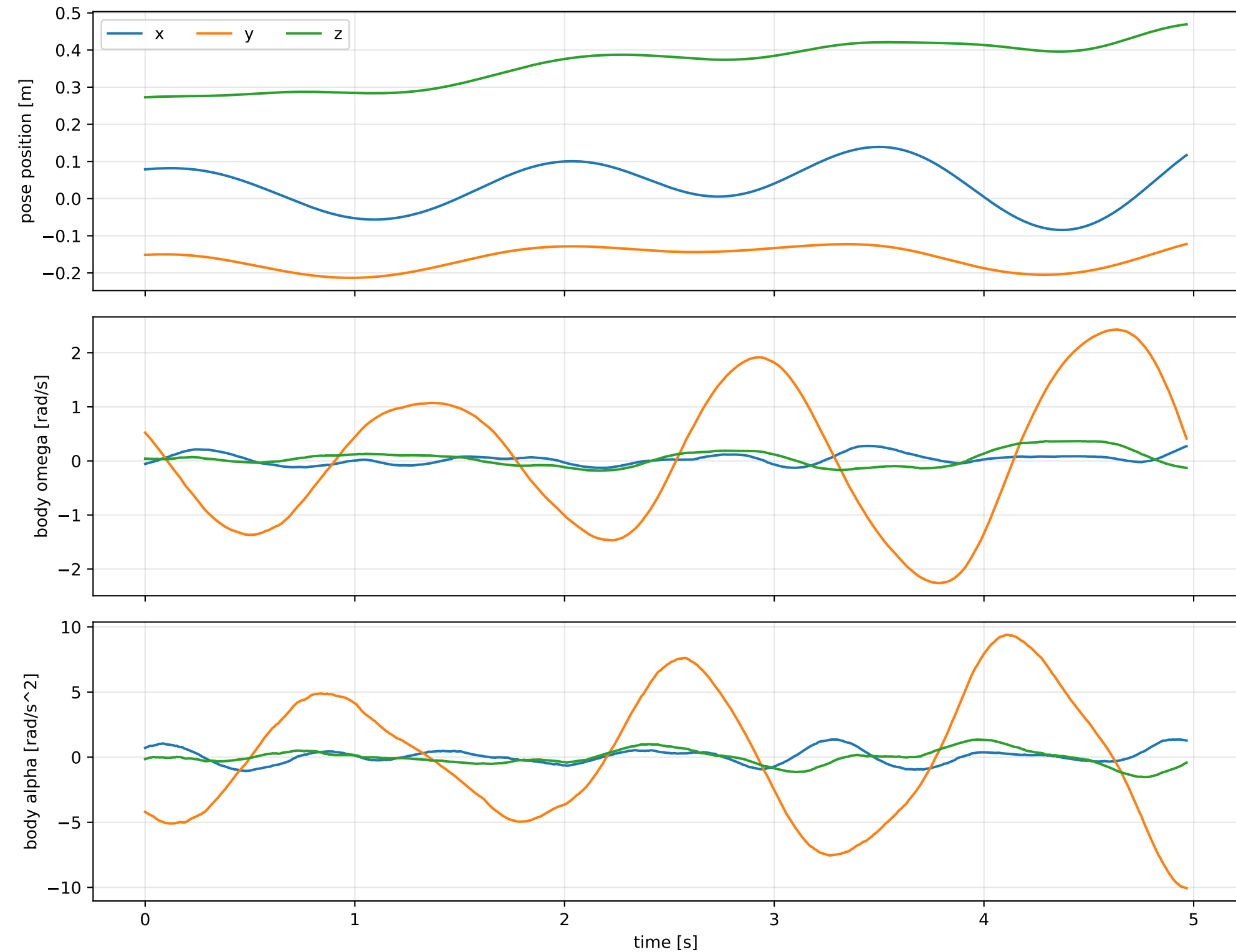
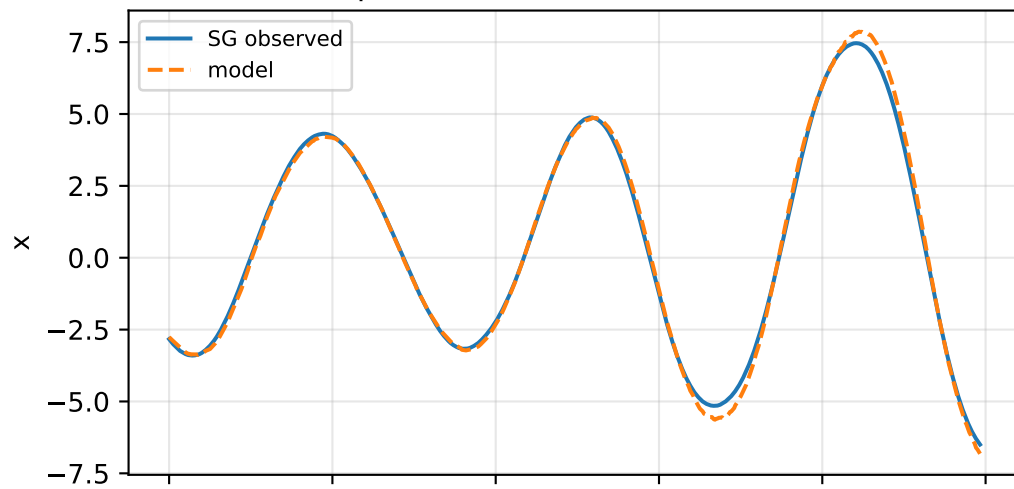


solver_standard_gauge_least_squares: SG trajectory and derived motion

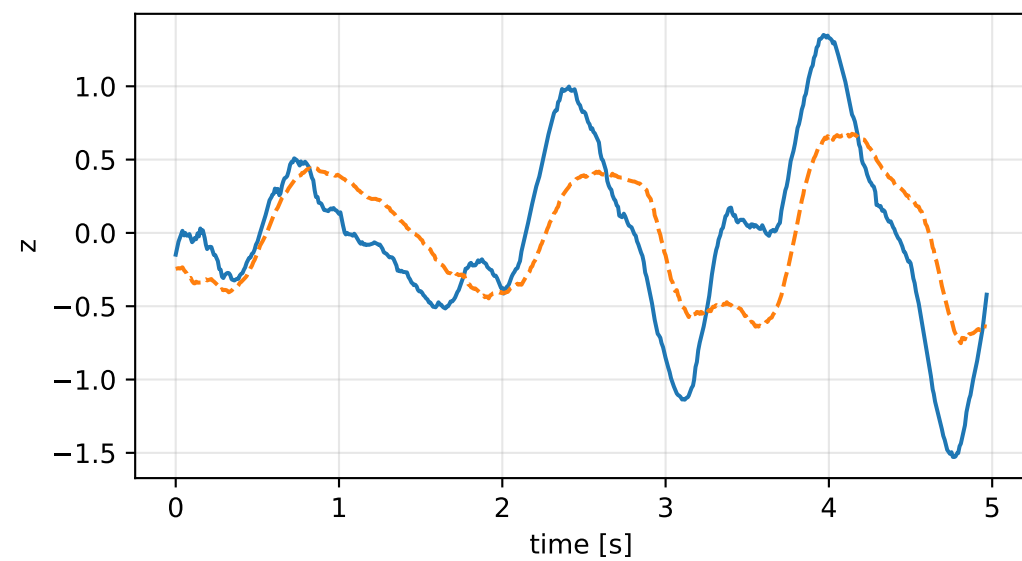
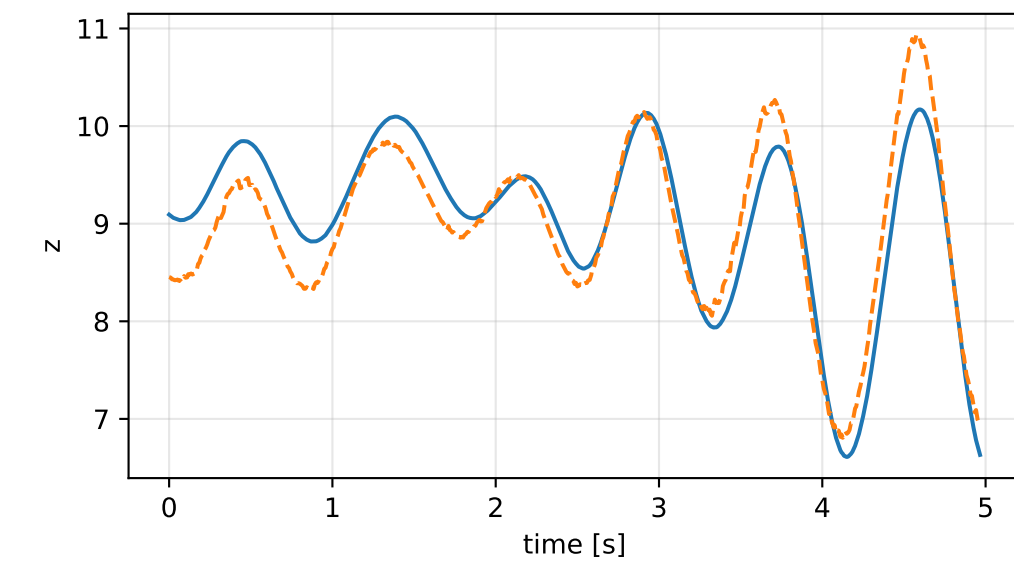
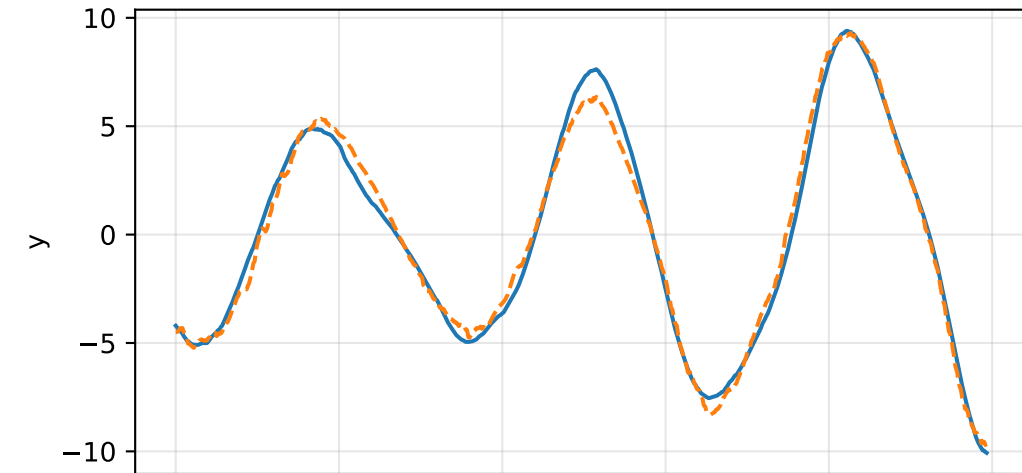
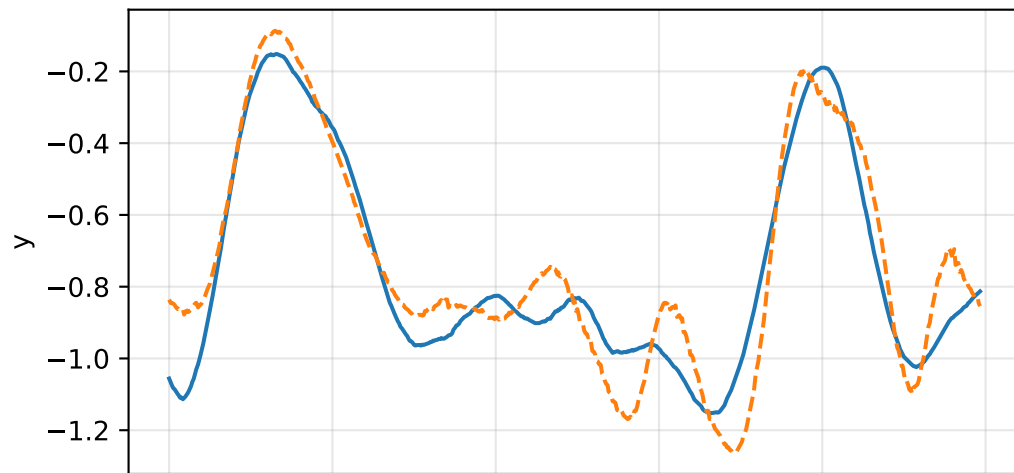
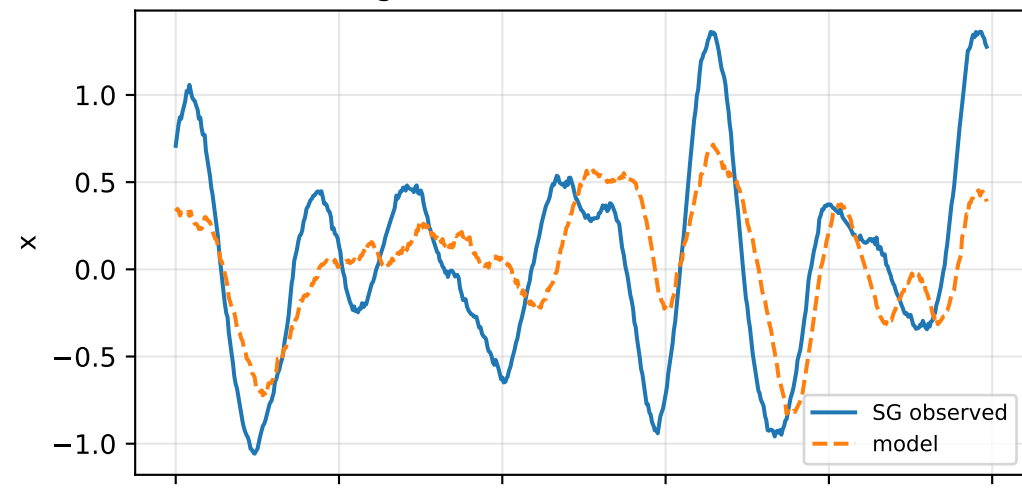


solver_standard_gauge_least_squares: acceleration residual objective

specific acceleration [m/s^2]

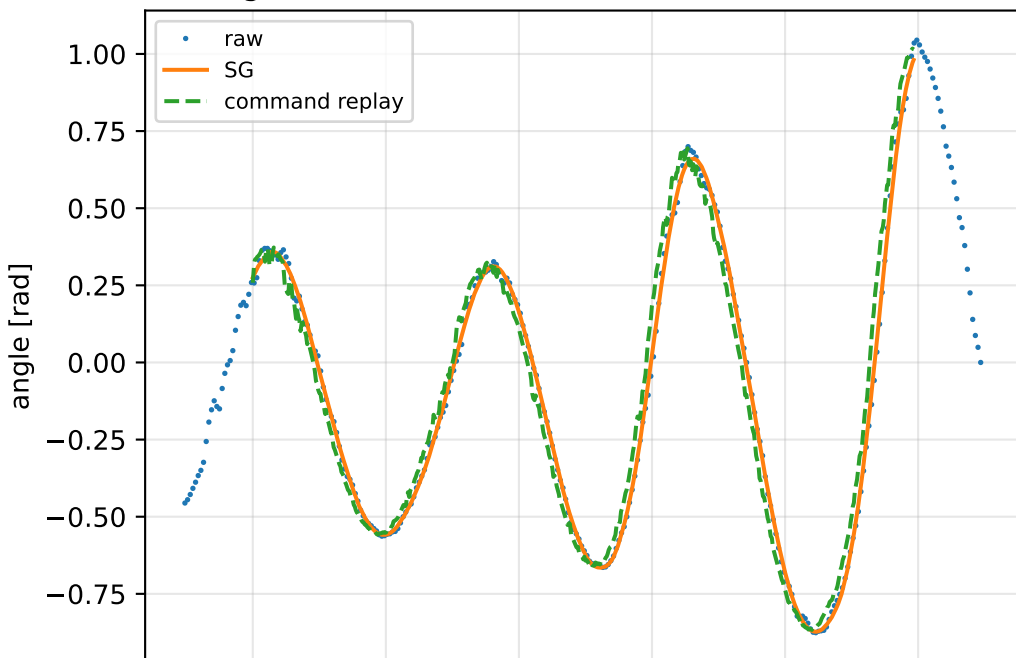


angular acceleration [rad/s^2]

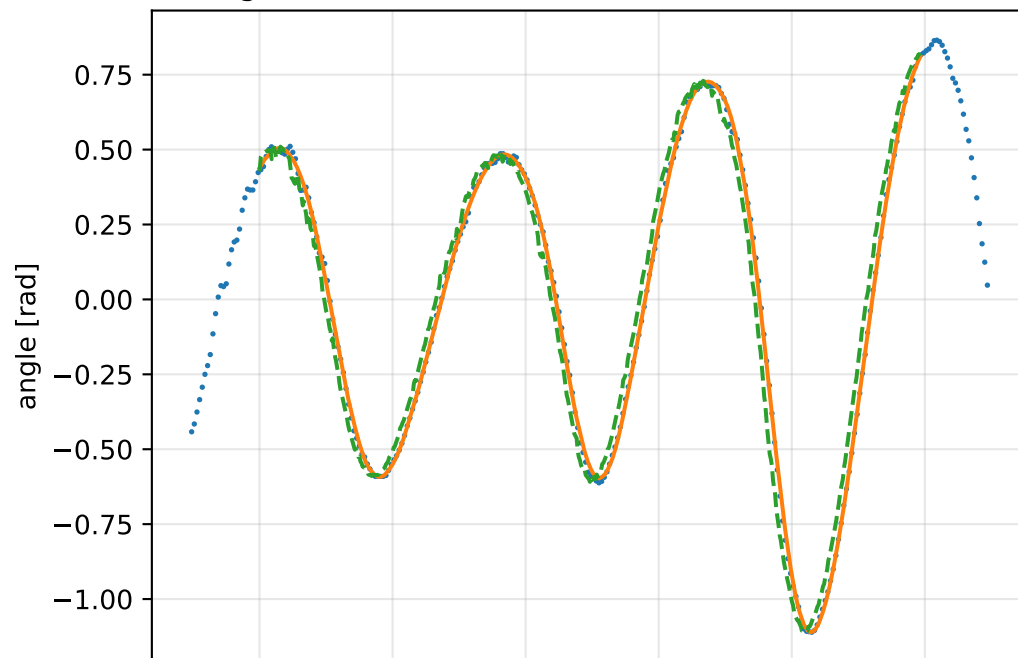


solver_standard_gauge_least_squares: gimbal measurement audit (objective=measured_sg)

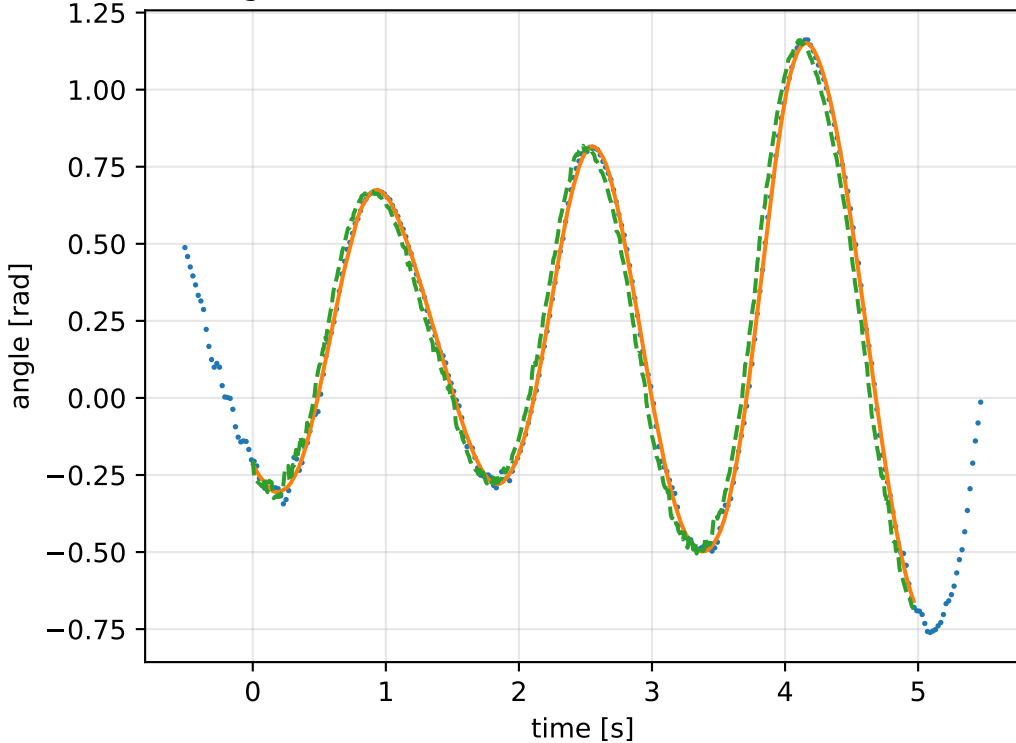
gimbal 1: RMSE=0.0759, max=0.2095 rad



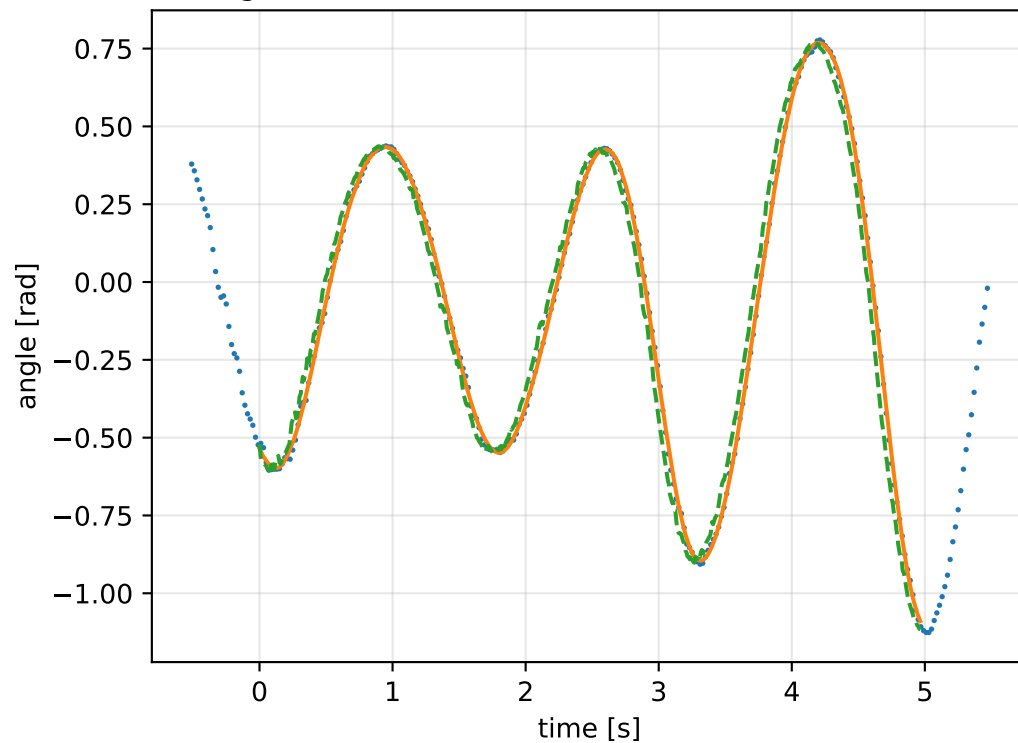
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

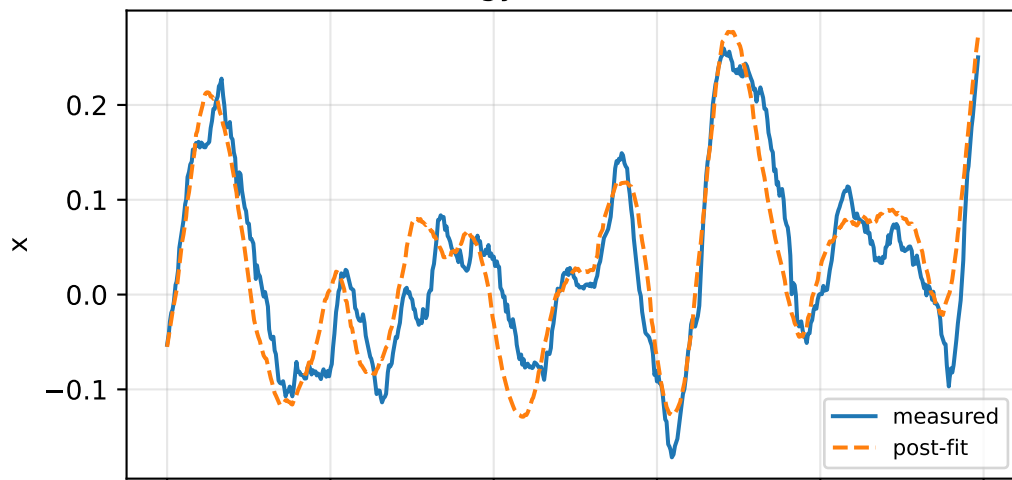


gimbal 4: RMSE=0.07131, max=0.1737 rad

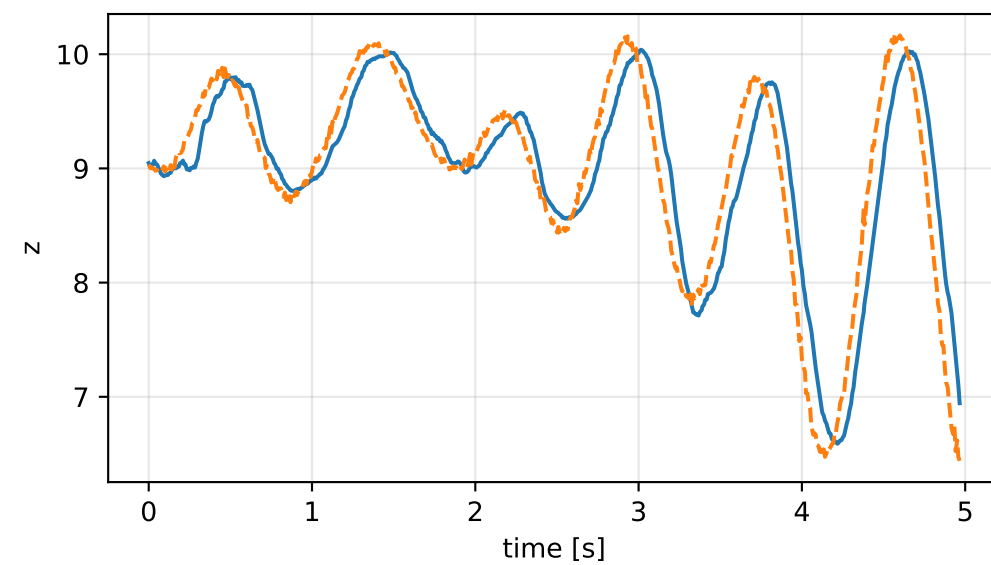
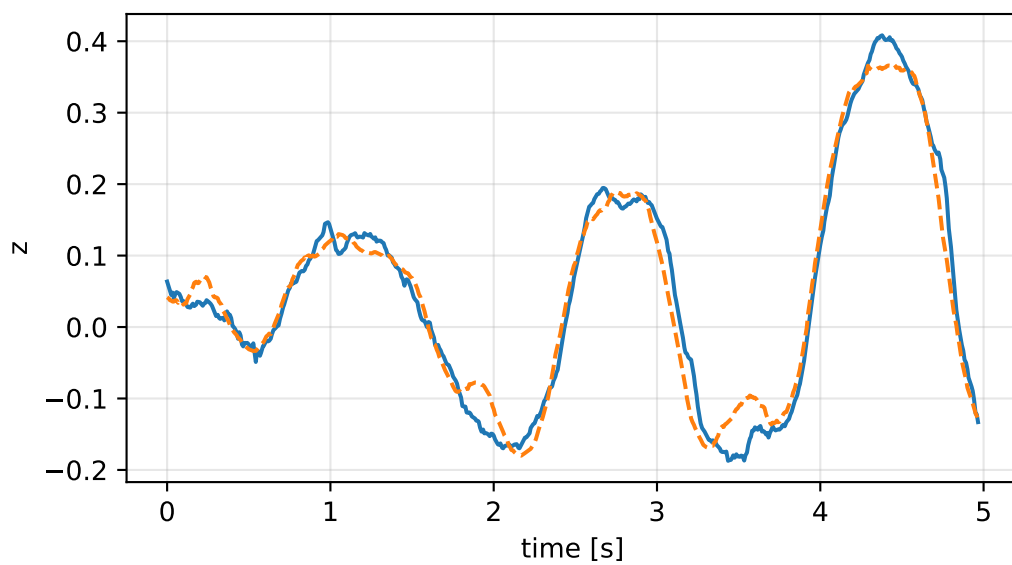
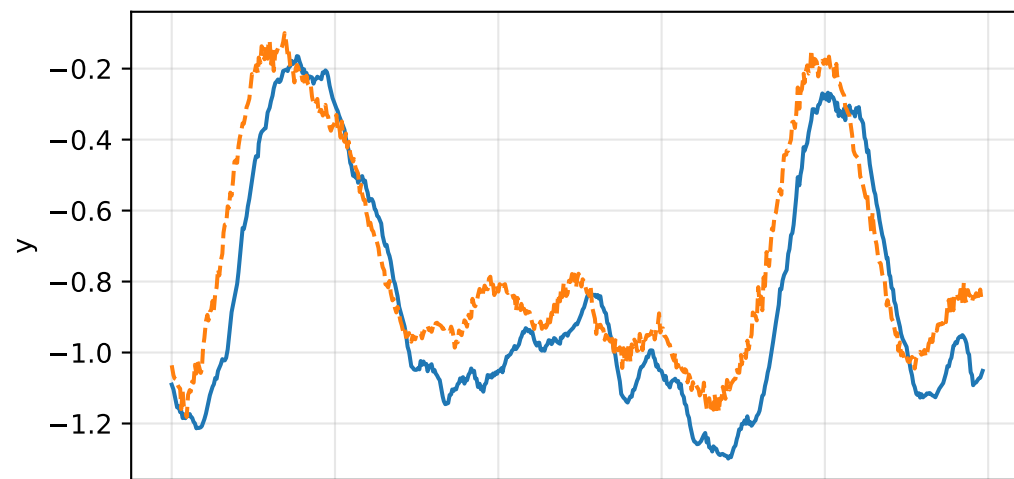
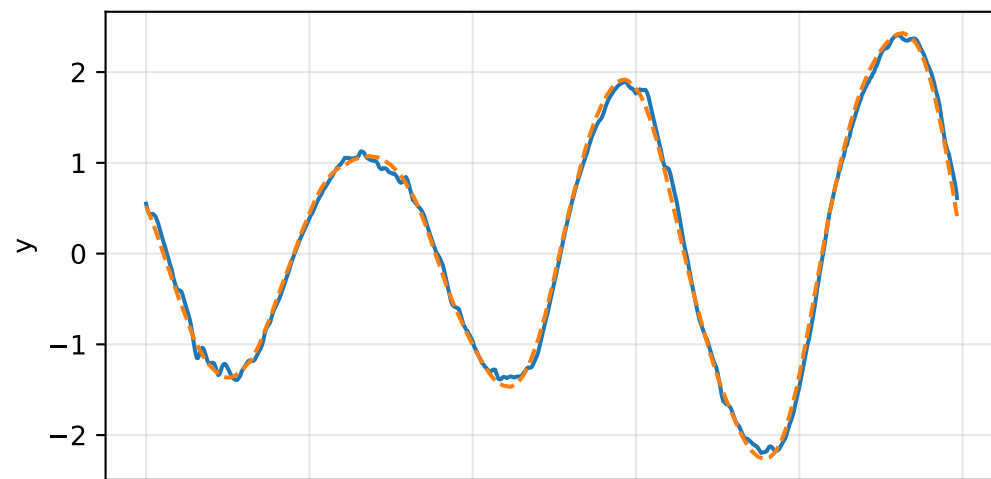
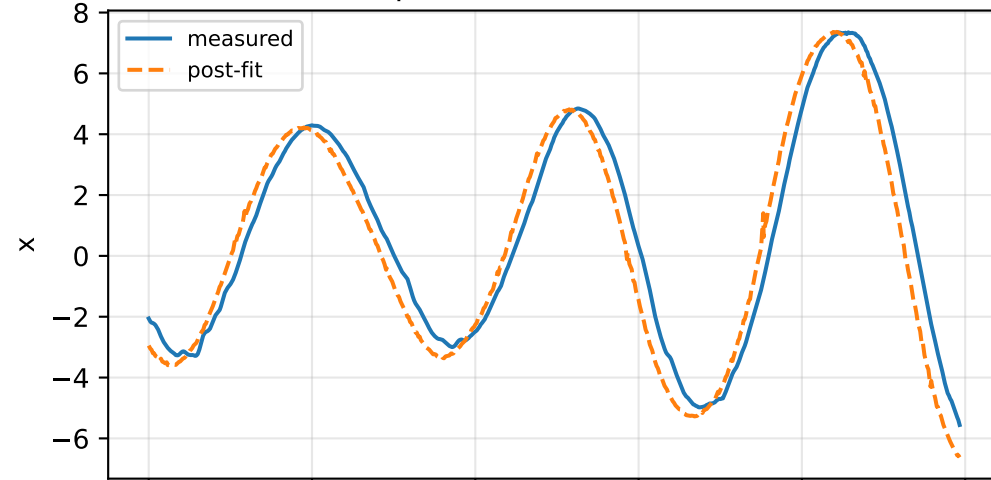


solver_standard_gauge_least_squares: IMU comparison (diagnostic only)

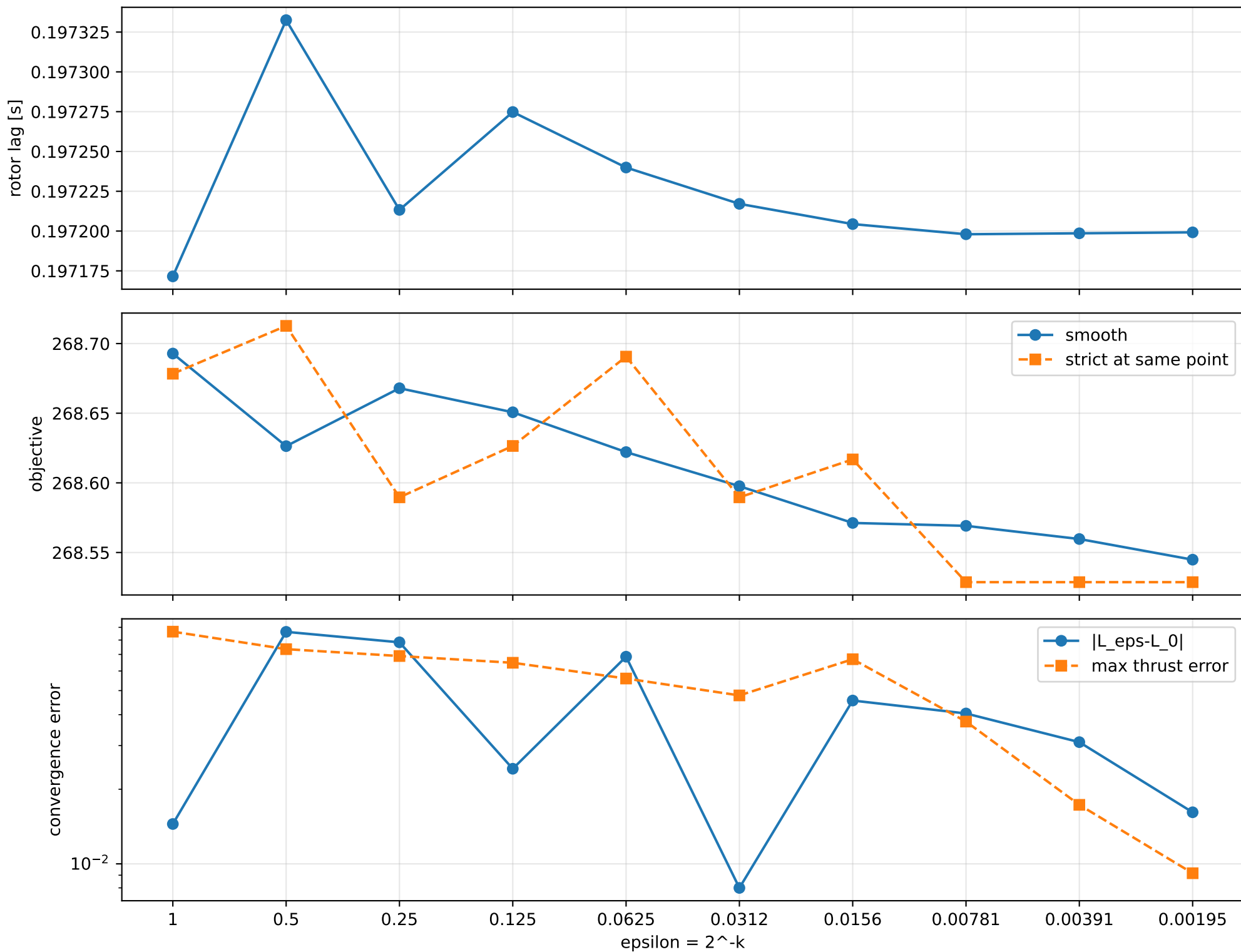
gyro [rad/s]



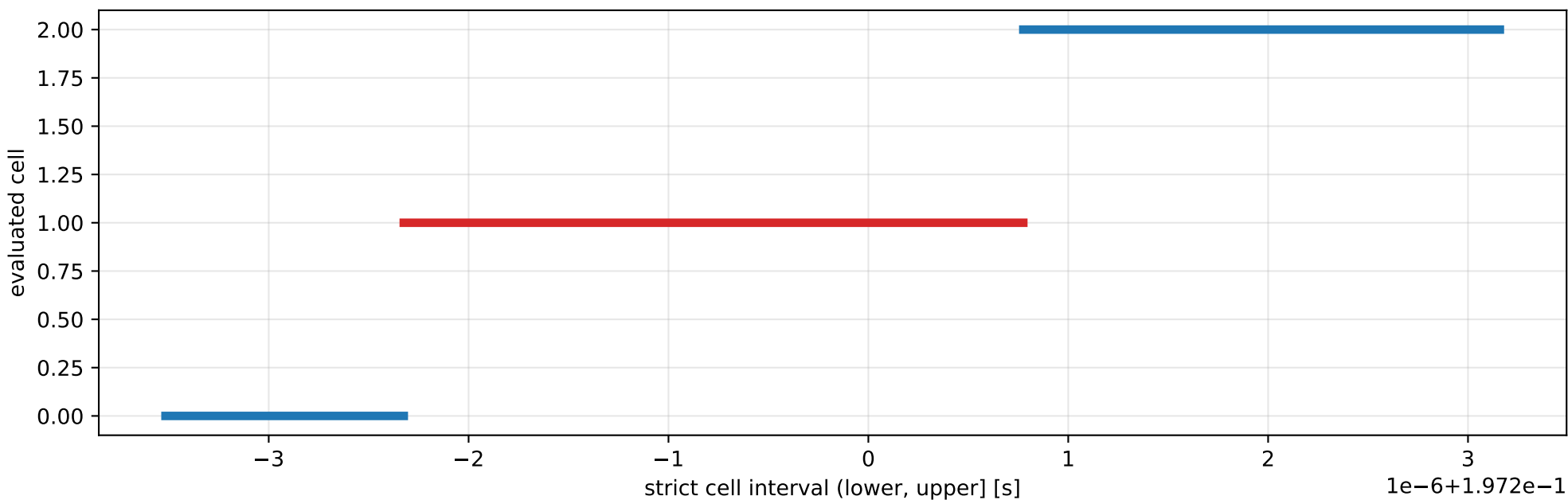
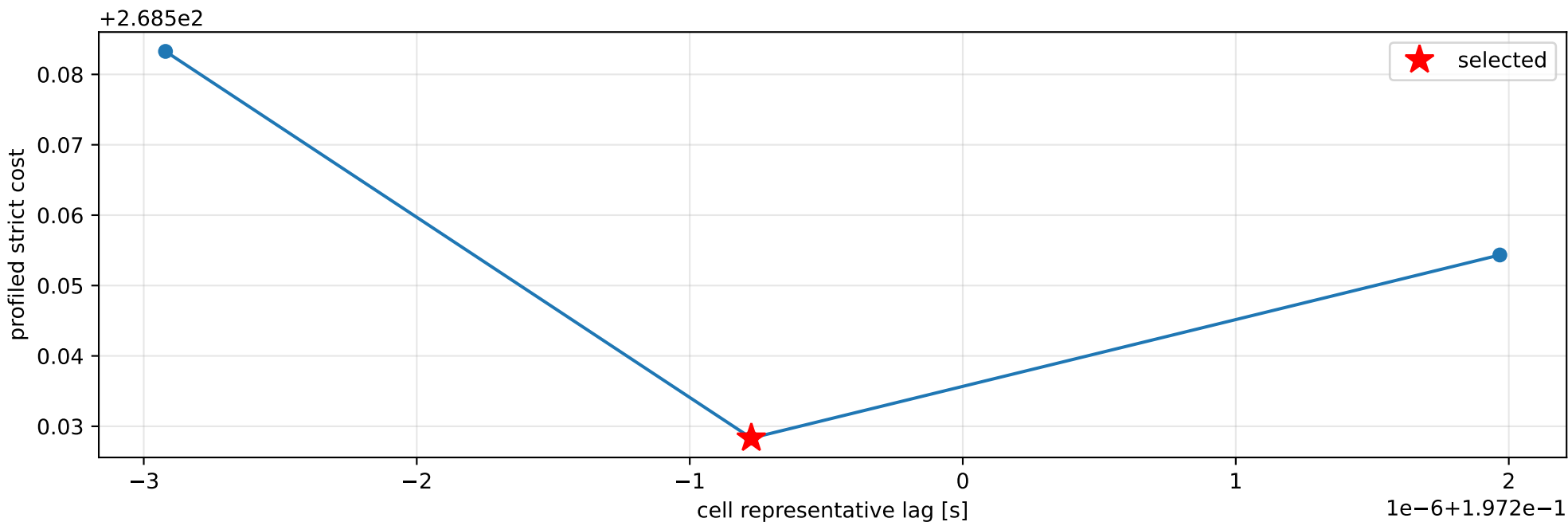
specific force [m/s^2]



solver_standard_gauge_least_squares: smooth-to-strict continuation



solver_standard_gauge_least_squares: exact strict-ZOH lag cells



selected (0.197197676, 0.197200775] s; final smooth 0.197199197 s; data boundary=False

solver_standard_gauge_least_squares: parameter estimate

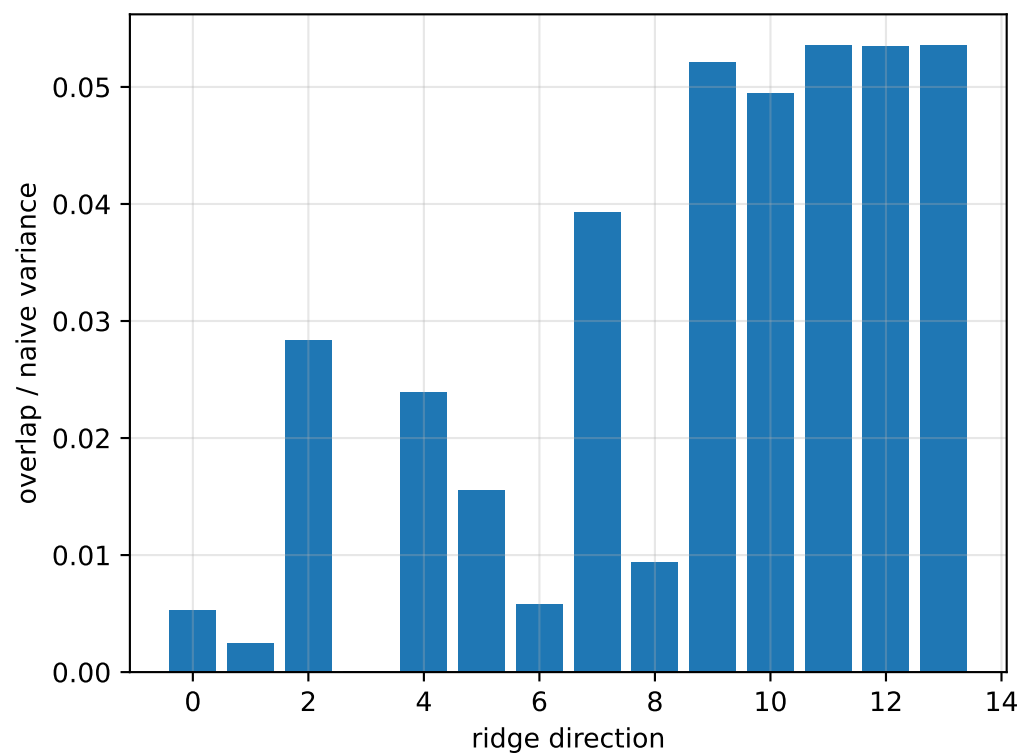
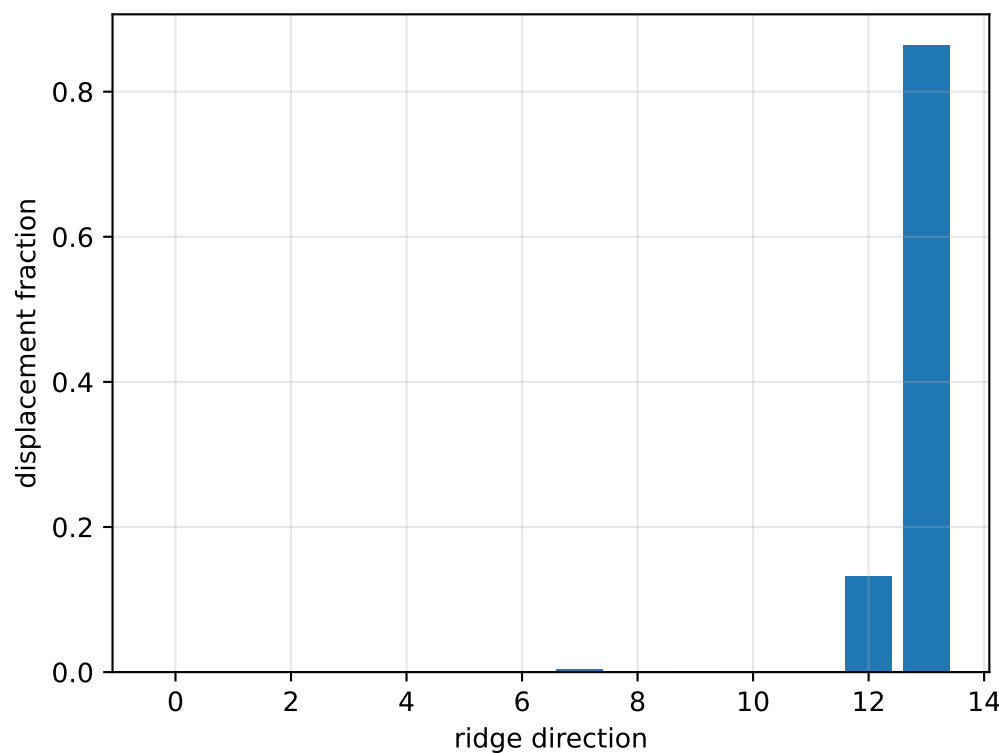
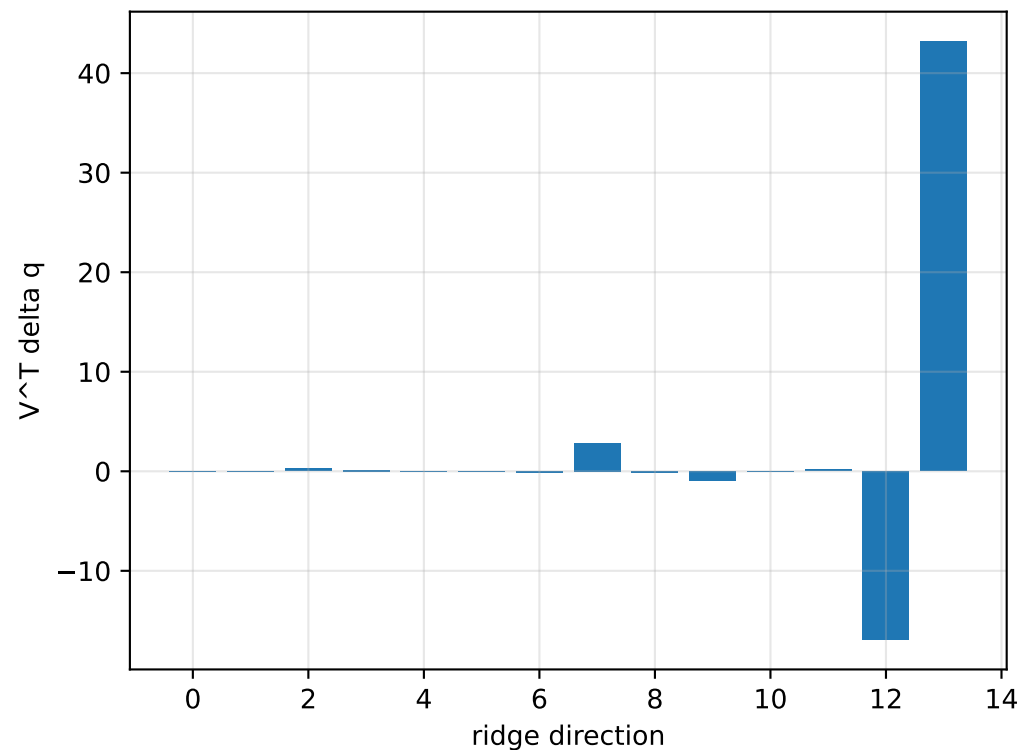
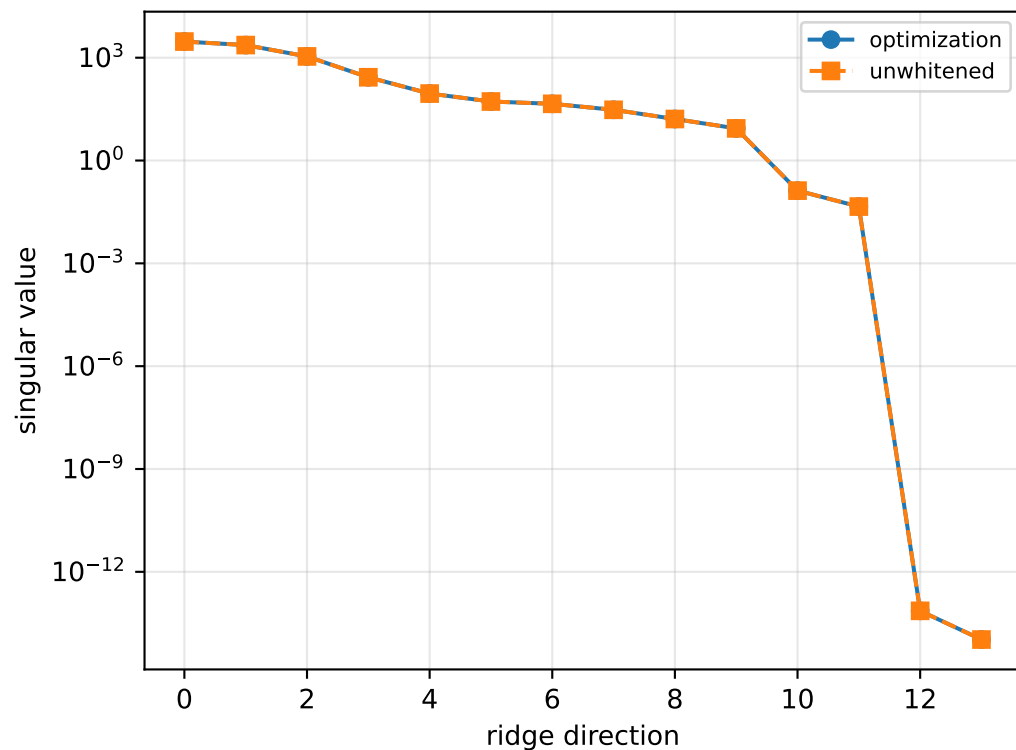
Case: solver_standard_gauge_least_squares
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 684.40911
inertia [kg m²]:
[[6.95373494e+01 -8.22249399e-02 -1.13092362e+01]
 [-8.22249399e-02 5.29431049e+01 8.67016608e-03]
 [-1.13092362e+01 8.67016608e-03 1.17405657e+02]]
CoG body [m]: [0.01762421 -0.00412196 -0.04437626]
force effectiveness: [229.22181708 295.16251485 257.73662121 213.70027587]

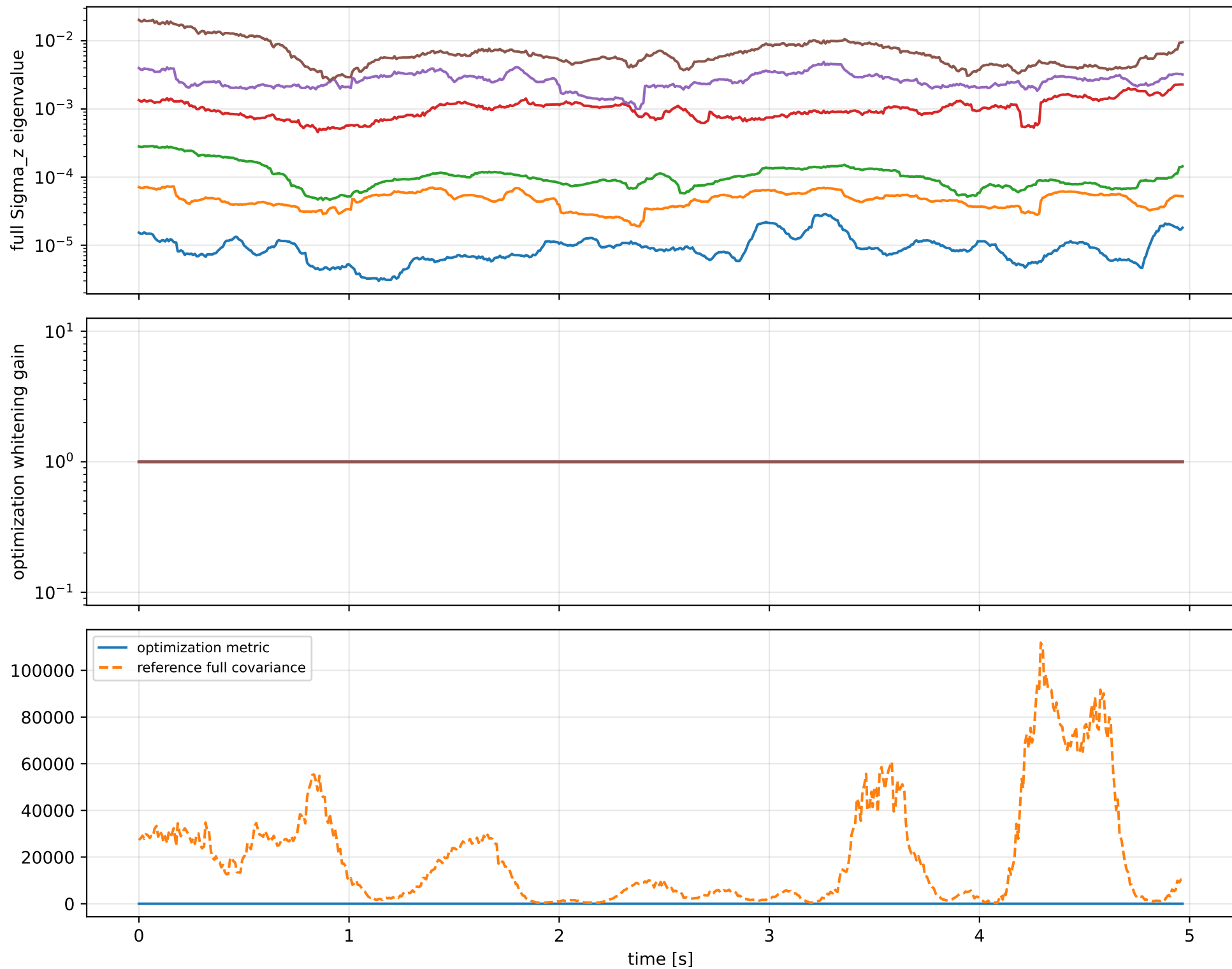
scale-free J/m [m²]:
[[1.01602022e-01 -1.20140043e-04 -1.65240878e-02]
 [-1.20140043e-04 7.73559325e-02 1.26681044e-05]
 [-1.65240878e-02 1.26681044e-05 1.71543095e-01]]
scale-free f/m: [0.3349193 0.4312662 0.37658269 0.31224055]

rotor lag cell: (0.197197676, 0.197200775] s
representative: 0.197199225 s

solver_standard_gauge_least_squares: ridge spectrum (rank 12, nullity 2, ||Jv||=4.82e-13)



solver_standard_gauge_least_squares: optimization vs reference covariance



solver_standard_gauge_least_squares: wrench / closure (force max 2.61e-12, torque max 1.25e-13)

